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B.Tech. III Semester (ECE/EIoT/ICE)

Roll no. \_\_\_\_\_

MID-SEMESTER EXAMINATION, SEPTEMBER 2024

Course Code: ECECC305/EIECC305/ICECC305  
Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 20

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

| Q.No. | Question                                                                                                                                                                                                                                                                                                                                                          | Marks | CO  |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| 1     | A network has four inputs ABCD and four outputs WXYZ. ABCD represents a binary coded decimal digit. Further, WX represents the quotient and YZ represents the remainder when ABCD is divided by 4 (WX and YZ represent 2 bit binary numbers). Assume that the invalid inputs do not occur.<br>(a) Draw the truth table.<br>(b) Implement by using NAND gates only | 2+2   | CO1 |
| 2     | The following Boolean function<br>$f(w, x, y, z) = [(w + \bar{x})y + \bar{y}\bar{z}]$<br>(a) Draw the truth table<br>(b) Implement it by using 8X1 MUX.                                                                                                                                                                                                           | 2+2   | CO2 |
| 3     | (a) Convert the number $(762)_8$ to 8421 code.<br>(b) Perform subtraction using 2's complement<br>$11011.001 - 1110.1101$                                                                                                                                                                                                                                         | 2+2   | CO1 |
| 4     | Simplify the following Boolean function $F$ with the don't-care conditions $d$<br>$F(A, B, C, D) = \Sigma (5, 6, 7, 12, 14, 15)$ and $d(A, B, C, D) = \Sigma (3, 9, 11)$<br>(a) Express in the SOP form<br>(b) Express in the POS form                                                                                                                            | 2+2   | CO2 |
| 5     | Given the function $f(A, B, C, D) = \Sigma_m (0, 4, 9, 10, 11, 12)$ and $f_1 = B \oplus D$ , and $f = f_1 f_2 + f_3$ . Determine the expressions of<br>(a) $f_2$<br>(b) $f_3$                                                                                                                                                                                     | 2+2   | CO1 |